Transport of Bisphenol A, Bisphenol S and three Bisphenol F isomers in saturated soils

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Abstract

With the limitation of the use of bisphenol A (BPA), the production of its substitutes, bisphenol S (BPS) and bisphenol F (4,4’-BPF) is increasing. Understanding the fate and transport of BPA and its substitutes in porous media can help reduce their risk of contaminating soil and groundwater systems. In this study, column and batch adsorption experiments were performed with ^{14}C-labeled bisphenol analogs and combined with mathematical models to investigate the interaction of BPA, BPS, 4,4’-BPF, 2,2’-BPF and 2,4’-BPF with four standard soils with different soil organic matter (SOM) contents. The results show that the transport capacity of BPS and 4,4’-BPF in the saturated soils is significantly stronger than that of BPA. Meanwhile, the mobility of the three isomers of bisphenol F (2,2’-BPF, 2,4’-BPF and 4,4’-BPF) showed some variability in saturated soils with high SOM content. The two-site kinetic retention mode was applied to simulate and interpret experimental data, and model simulations described the interactions between the bisphenol analogs and soil very well. The fitting results show that SOM provides more adsorption sites for bisphenol analogs and these adsorption sites may be irreversible adsorption sites. For the different mobility of bisphenol analogs, hydrophobicity is the main factor leading to the difference in adsorption affinity between BPA, BPS, 4,4’-BPF and soil. The main factor leading to the difference of adsorption affinity between 4,4’-BPF and its isomers (2,2’-BPF and 2,4’-BPF) and soil may be hydrogen bonding force. In addition, the results of this study show that the relatively high mobility of BPA substitutes BPS and 4,4’-BPF may pose a significant risk to groundwater quality, so 4,4’-BPF and BPS may not be environmentally friendly alternatives to BPA. In addition, as by-products of 4,4’-BPF production, 2,2’-BPF and 2,4’-BPF have high mobility in soil and may pose a more significant threat to groundwater than 4,4’-BPF.

Highlights

- BPS and 4,4’-BPF has higher mobility than BPA in four standard soils.
- 2,2’-BPF, 2,4’-BPF and 4,4’-BPF have different transport capacities in standard soils.
- The adsorption affinity of bisphenol analogs to soil is significantly correlated with soil organic matter.

1. Introduction

2,2-Bis(4-hydroxyphenyl) propane (BPA) is a derivative of phenol and acetone (Rochester, 2013; Staples et al., 1998). It is widely used in the manufacture of important chemical raw materials such as polycarbonate and epoxy resin (Catenza et al., 2021; Huang et al., 2012; Ma et al., 2019). It is worth noting that in the manufacturing process of plastic products, the addition of bisphenol A can make it colorless and transparent, durable, lightweight and outstanding impact resistance, so bisphenol A has been widely used in plastic products, milk bottles, canned food coating and so on (Vilarinho et al., 2019). Due to the practicability of BPA, the annual output of BPA reached 5 million tons in 2015. However, studies have found that BPA is an endocrine disruptor (EDC) which disrupts the balance of the endocrine system in animals and humans (Akash et al., 2020; vom Saal et al., 2012). Many countries have started to restrict the use of BPA and there is an urgent need to develop and use alternatives to BPA (Catenza et al., 2021; Chen et al., 2016; Rochester, 2013). 4-hydroxyphenylsulfone (BPS) and 4-methylene-4-methylenediphenol (4,4’-BPF) are the main substitutes for BPA. BPS is mainly used in the production of color-fixing agents, resin flame retardants and so on, as well as the intermediate of pesticides and dyes (Liu et al., 2021). BPF is commonly used in the synthesis of flame retardants, antioxidants and indicative active agents (Qian et al., 2021). BPF used in industry usually refers to 4,4'-BPF, but in the production of BPF, it is inevitable to produce two kinds of isomers: 2,2'-BPF and 2,4'-BPF. However, few people pay attention to 2,2'-BPF and 2,4'-BPF (Guo et al., 2019; Sun et al., 2017). Due to the structural similarity between 4,4'-BPF and BPS and BPA, BPS and 4,4'-BPF were found to have the same potency in inhibitory hormone signaling as BPA (Gogola–Mruk et al., 2023; Le Fol et al., 2017; Wang et al., 2021). The high use of bisphenol analogs has led to their high detection frequency in various environmental media (Gao et al., 2023; Maturi et al., 2023; Qian et al., 2021; Xue et al., 2016). Among these, the soil environment has been found to be an important sink for bisphenol analogs (Xu et al., 2021). Bisphenol analogs in the soil can enter the aqueous environment through a variety of pathways, such as surface runoff, drainage flow and groundwater flow (Dueñas–Moreno et al., 2022). Due to unfavorable redox and degradation conditions in the groundwater, contaminants degrade very slowly (Yamazaki et al., 2015; Zhi et al., 2019). Once the contaminants reach the groundwater, the risk to human health is high. Therefore, understanding the mechanisms that control the transport of BPA and its substitutes through soil is essential for developing management practices to protect groundwater.

Several studies have shown that BPA, as a polar contaminant, is highly mobile and is likely to be able to cross the soil profile and enter groundwater. For example, (Dai et al., 2020) reported that BPA is highly mobile in porous media in both saturated and unsaturated conditions, and almost all of it flows out of the soil column in soils with low SOM content. The transport capacity of BPA in soil seems to be closely related to the rate of pore water in the soil, soil pH and SOM content (Guo et al., 2022; Zakari et al., 2016). Notably, as an alternative, BPS seems to have higher mobility than BPA (Shi et al., 2019; Shi et al., 2018). Adsorption is a key process to control the transport and fate of organic pollutants in various environments. BPA seems to have a stronger affinity for soil adsorption when in a non-dissociated state ($pH < pK_a$) due to its greater hydrophobicity than BPS (Choi &Lee, 2017). This makes BPA more inclined to deposit in the soil and more BPS inclined to flow into the groundwater. However, the survey shows that the concentration of BPF seems to be one to two orders of magnitude higher than BPA in Japanese, Korean and Chinese waters (Qian et al., 2021; Yamazaki et al., 2015). Current research in BPF seems to be limited to the development of new acid catalysts and optimization of BPF isomer product distribution (Sun et al., 2017; Wang et al., 2016). The only few studies on BPF retention and distribution have focused on 4,4'- BPF (Liu et al., 2018; Wu et al., 2019). There is a gap in studies on the adsorption and transport of 2,2'-BPF, 2,4'-BPF in soils. Moreover, the transport of bisphenol analogs in different types of soils thus might be differ, this has not been tested.

The overarching goal of this study is to advance the current understanding of the fate and transport of bisphenol analogs in soil. Five bisphenol analogs including BPA, BPS, 4,4'-BPF, 2,2'-BPF, 2,4'-BPF uses ^{14}C-labeled for column experiments and batch adsorption experiments. Four standard soils from Germany were used as media. The standard soils are available in large quantities and are representative and ecologically relevant for generating reproducible and comparable data (Bastos et al., 2014; Hofman et al., 2009). The use of ^{14}C-labeled bisphenol analogs and standard soils for column experiments and batch adsorption experiments can effectively improve the accuracy and reproducibility of experimental data (Schaefer, 1991). Mathematical models are used to simulate and interpret experimental data. The objectives are as follows: 1) To compare the transport capacity of BPA and its substitutes, BPS and 4,4'-BPF in different soils, and to reveal the mechanism of interaction with soil. 2) The transport capacity of 4,4’-BPF and its isomers 2,2’-BPF and 2,4’-BPF in different soils was compared, and the interaction mechanism with soil was revealed. 3) To establish and test mathematical models for fate and transport of bisphenol analogizes in porous media.

2. Materials and methods

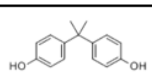
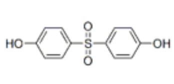
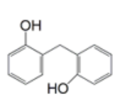
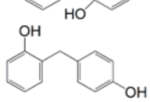
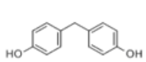
2.1 Materials

Lufa soil was purchased from Lufa Speyer (Speyer, Germany). The specific properties of the soil are shown in Table 1. BPA, BPS, 4,4’-BPF, 2,4’-BPF and 2,2’-BPF (with purity > 99%) was purchased from Sigma – Aldrich (St. Louis, MO). The physicochemical characteristics of the bisphenol analogs are shown in Table 2.

Table 1
Properties of Lufa soils.

Soil	Soil type (USDA)	Sand	Silt	Clay	Organic carbon in % C	pH	CEC (meq / 100g)	Weight per volume (g L^{-1})
Lufa a	sand	86 ± 0.9	11.5 ± 1.2	2.5 ± 0.7	1.23 ± 0.10	4.9 ± 0.3	4.2 ± 0.6	1436 ± 41
Lufa b	sandy loam	76.1 ± 4	16.2 ± 2.6	7.7 ± 1.7	2.94 ± 0.25	6.0 ± 0.2	9.7 ± 0.4	1218 ± 47
Lufa c	loam	33.2 ± 2.1	41.1 ± 1.2	25.8 ± 1.8	1.84 ± 0.32	7.3 ± 0.1	33.0 ± 4.5	1265 ± 38
Lufa d	sandy loam	58.2 ± 1.6	30.9 ± 1.4	10.9 ± 0.9	1.01 ± 0.09	7.3 ± 0.1	17.4 ± 3.6	1264 ± 72

Table 2. Physicochemical properties of contaminants.

Contaminant	Chemical structure	MW (g/mol)	Log K_{OW}	pK_a	References
BPA		228.29	3.32	pK_{a1}=9.78, pK_{a2}=10.39	(Choi &Lee, 2017)
BPS		250.27	1.65	pK_{a1}=7.42, pK_{a2}=8.03	(Choi &Lee, 2017)
2,2’-BPF		200.24	2.73	pK_{a1}=8.26, pK_{a2}=10.1	(Škufca et al., 2021)
2,4’-BPF		200.24	2.73	pK_{a1}=9.8, pK_{a2}=10.44	(Škufca et al., 2021)
4,4’-BPF		200.24	2.73	pK_{a1}=9.84 pK_{a2}=10.45	(Škufca et al., 2021)

2.2 Determination of pollutant concentration

^{14}C-labeled BPA (0.74 GBq/mmol), BPS (2.40 GBq/mmol), 2,2’-BPF (2.82 GBq/mmol), 2,4’-BPF (2.82 GBq/mmol) and 4,4’-BPF (2.82 GBq/mmol) were synthesized using ^{14}C-labeled phenol as a precursor. Liquid scintillation counter (LS6500, Beckman Coulter, Fullerton, CA) was used to measure the radioactive response value (DPM) of ^{14}C labeled pollutants to quantify the mass of pollutants. Combined with weighing, the volume of samples and the concentration of pollutants were calculated.

2.3 Column Transport Experiments

Column experiments were performed in a 10 cm long borosilicate glass column apparatus with a core diameter of 0.66 cm. air-dried, screened Lufa soil was uniformly filled in the borosilicate glass column. Carbon dioxide gas (purity > 99%) was introduced into the soil column from the lower part of the end soil column for about 1 h. The residual air in the column was exhausted. Then, deionized (DI) water was pumped from the bottom of the soil column to the bottom to flush the filling medium. Subsequently, approximately 200 mL of background electrolyte (0.5 mM NaCl solution) was injected from bottom to top by an injection pump to flush the soil column to equilibrium. When the soil column is equilibrated by long-term saturation with the background solution, the injection solution is injected into the soil column at a controlled flow rate of 10 m/d according to the experimental setup, and an effluent sample is collected every 2–3 well volumes (PV). After collecting 60 PV (contaminants reached migration equilibrium in the soil column), the column was flushed with background solution (free of contaminants) and an effluent sample was collected every 2–3 PV until no contaminants were detected in the effluent. The effluent samples were measured with a liquid scintillation counter. The basic parameters of the soil column in the test are shown in Table 3.

Table 3. Experimental setups and breakthrough results of BPA, BPS, 4,4’-BPF, 2,2’-BPF and 2,4’-BPF transport experiments.

No.	Column properties				Influent properties			Effluent properties [a]
	Soil	Length (cm)	Bulk density (g/cm^3)	Porosity (-)	Contaminant	Contaminant concn. (μg/L)	pH	$C/C_{0-\text{cont.}}$ (%)
1	Lufa a	7.08	1.48	0.44	BPA	10.02	6.7	77.5 ± 1.2
2	Lufa a	6.85	1.49	0.44	BPS	9.93	6.5	94.1 ± 1.1
3	Lufa a	7.18	1.51	0.43	2,2'-BPF	9.61	6.8	94.4 ± 0.7
4	Lufa a	7.20	1.49	0.44	2,4'-BPF	10.08	6.7	89.6 ± 1.4
5	Lufa a	7.14	1.52	0.43	4,4'-BPF	9.68	6.8	92.8 ± 2.0
6	Lufa b	7.18	1.39	0.48	BPA	10.04	6.5	51.6 ± 0.1
7	Lufa b	7.21	1.39	0.48	BPS	9.93	6.7	90.8 ± 0.9
8	Lufa b	6.94	1.41	0.47	2,2'-BPF	10.52	6.6	92.7 ± 1.5
9	Lufa b	6.90	1.39	0.48	2,4'-BPF	9.43	6.5	89.3 ± 2.3
10	Lufa b	7.01	1.39	0.48	4,4'-BPF	10.23	6.7	76.0 ± 0.7
11	Lufa c	6.95	1.37	0.48	BPA	9.55	6.8	67.3 ± 0.4
12	Lufa c	6.78	1.34	0.49	BPS	9.96	6.7	97.1 ± 1.5
13	Lufa c	6.79	1.35	0.49	2,2'-BPF	9.56	6.6	100.6 ± 2.4
14	Lufa c	6.86	1.37	0.48	2,4'-BPF	9.87	6.5	92.3 ± 0.8
15	Lufa c	6.95	1.39	0.48	4,4'-BPF	10.22	6.9	93.5 ± 1.1
16	Lufa d	6.98	1.36	0.49	BPA	9.88	6.8	74.5 ± 0.5
17	Lufa d	6.90	1.35	0.49	BPS	9.92	6.7	101.3 ± 0.8

18	Lufa d	6.98	1.34	0.49	2,2’-BPF	9.75	6.7	97.7 ± 1.2
19	Lufa d	6.78	1.40	0.47	2,4’-BPF	9.21	6.6	96.9 ± 0.1
20	Lufa d	6.94	1.37	0.48	4,4’-BPF	9.91	6.8	96.9 ± 0.8

2.4 Two-site kinetic retention model

The BTCs of soil BPs were simulated using the one-dimensional CDE with kinetic retention by two site types. The two-site kinetic retention model interprets the mass transfer of soil bisphenol analogs between the aqueous and solid phase: at the first site (S_1) reversible retention is assumed, whereas at the second site (S_2) the retention is irreversible. The CDE in this case can be expressed as:

$$\frac{\partial \theta C}{\partial t} + \rho_b \frac{\partial (S_1)}{\partial t} + \rho_b \frac{\partial (S_2)}{\partial t} = \frac{\partial}{\partial x}(\theta D \frac{\partial C}{\partial x}) - \frac{\partial q C}{\partial x} \quad (3)$$

$$\rho_b \frac{\partial (S_1)}{\partial t} = \theta k_1 C - \rho_b k_{1d} S_1 \quad (4)$$

$$\rho_b \frac{\partial (S_2)}{\partial t} = \theta k_2 \Psi_t C_{\mathrm{t}} \quad (5)$$

where θ is the volumetric water content, C is the soil bisphenol analogs concentration in the aqueous phase [NL^{-3}], t is the time [T], is the bulk density of the Lufa soil [ML^{-3}], where M denotes the unit of mass, x is the vertical spatial coordinate [L], D is the hydrodynamic dispersion coefficient [L_2T^{-1}], q is the Darcy velocity [LT^{-1}], and S_1 [NM^{-1}] and S_2 [NM^{-1}] are the solid-phase concentrations associated with retention on site 1 and 2, respectively. k_1 [T^{-1}] and k_2 [T^{-1}] are the first-order adsorption coefficients for site 1 and 2, respectively, k_{1d} [T^{-1}] is the first-order desorption coefficient.

2.5 Batch Adsorption Experiments

The adsorption experiment of organic pollutants on Lufa soil was carried out by the adsorption experiment method used in the literature. First, about 0.5 g of soil was added to a series of 20 mL brown glass bottles; then about 20 mL of electrolyte solution containing 0.5 mM NaCl was added and wetted for 24 hours. After that, different amounts of ^{14}C of labeled organic pollutant reserve liquid were carefully injected into the glass bottle with a micro syringe. The bottle was then sealed with a polytetrafluoroethylene pad and cap and placed on a rotating disk in a darkroom that ran continuously for seven days at a rate of three cycles per minute to equalize the adsorption of organic pollutants on the soil. After balancing the adsorption, measurement of pollutant concentration.

Freundlich model and Linear model were used to fit the adsorption isotherm of bisphenol analogs in Lufa soil. The calculation formulas of the two models are as follows:

Freundlich model (FM):

$$q_e = K_F C_e^{\frac{1}{n}} \tag{1}$$

Linear model (LM):

$$q_e = K_d C_e \tag{2}$$

Where: q_e (mg / kg) is the adsorption state concentration of organic pollutants after adsorption equilibrium, C_e (mg / L) is the aqueous phase concentration of organic pollutants after adsorption equilibrium, K_F ((mg / kg) / (mg / L)) is the Freundlich adsorption coefficient, n (unitless) is the Freundlich index, K_d (L/kg) is the Lufa water partition coefficient.

3. Results and discussion

3.1 Transport of BPA and its substitutes in saturated Lufa soils

In all four typical saturated soil media, the BPA substitutes exhibited stronger transport than BPA. Figure 1 shows the measurement and fitting breakthrough curves (BTCs) of BPA, BPS and 4,4'-BPF columns filled with four different types of standard soils under saturated flow conditions. Figure 1 shows that the maximum penetration rates (C/C_0) of BPS and 4,4'-BPF in four typical soils were 89.3-101.3%, while the maximum penetration rates (C/C_0) of BPA in four soils were 51.6–77.5%. In addition, the differences in mobility of the three bisphenol analogs in the four typical soils seemed to become significant with increasing SOM content. The SOM content of four Lufa soils follows the order of Lufa b (2.94%) > Lufa c (1.84%) > Lufa a (1.23%) > Lufa d (1.01%). The higher SOM content in Lufa b seems to make the difference in mobility of BPA and its substitutes more significant. This may be caused by the fact that SOM provides more adsorption sites for BPA and its substitutes, while the three bisphenol analogs have different binding abilities to SOM. The effect of SOM content on the transport of bisphenol analogs in porous media has been reported in other studies [Chen et al., 2022; Dai et al., 2020; Shi et al., 2019]. For example, [Shi et al., 2019] in the study, bisphenol A and bisphenol S had low mobility in soils with high SOM content, but their transport in the sand column was significantly enhanced after removal of SOM from the soil. The two-site kinetic retention model matched well with the experimental BTC of BPA, BPS and 4,4'-BPF in the four soils (Fig. 1). All fitted parameters are shown in Table 5, and R^2 > 0.96 for all fitted results. The fitted results indicate that the adsorption coefficient k_1 for reversible adsorption site 1 is greater than that for irreversible adsorption site k_2, indicating that more reversible than irreversible adsorption of bisphenol analogue occurs in the soils due to the stronger mobility. Among them, the k_2 values representing the irreversible sorption coefficients of the bisphenol analogs in the four soils were positively correlated with the content of SOM in the soil. Meanwhile, the k_2 values were negatively correlated with the permeability of BPA, BPS and 4,4'-BPF, and the k_2 values of BPA and its substitutes in the four soils were BPA > BPA substitutes. This confirms that SOM provides more adsorption sites for BPA, BPS and 4,4'-BPF and that BPA binds more strongly to these adsorption sites than its alternatives.

Table 4. Adsorption parameters of BPA, BPS, 2,2'-BPF, 2,4'-BPF and 4,4'-BPF on Lufa soils.

Contaminant	Soil	K_F (L/kg)	*1/n*	R^2	K_d (L/kg)	R^2
BPA	Lufa a	131.2838	1.2023	0.9723	56.216	0.9540
BPS	Lufa a	12.0443	0.9368	0.9893	15.097	0.9822
2,2'-BPF	Lufa a	21.7878	0.9489	0.9999	26.321	0.9863
2,4'-BPF	Lufa a	58.7120	1.2358	0.9916	24.192	0.9692
4,4'-BPF	Lufa a	14.6867	0.9275	0.9802	19.137	0.9702
BPA	Lufa b	63.6444	1.0125	0.9960	60.318	0.9904
BPS	Lufa b	16.8033	0.9087	0.9748	23.571	0.9681
2,2'-BPF	Lufa b	48.3489	1.0771	0.9965	35.704	0.9853
2,4'-BPF	Lufa b	43.5763	1.0450	0.9960	36.510	0.9914
4,4'-BPF	Lufa b	80.5778	1.1898	0.9990	37.978	0.9850
BPA	Lufa c	80.0722	1.0979	0.9977	53.524	0.9936
BPS	Lufa c	12.2702	0.9686	0.9874	13.723	0.9854
2,2'-BPF	Lufa c	31.6415	1.0220	0.9950	29.057	0.9841
2,4'-BPF	Lufa c	47.9836	1.0882	0.9989	33.962	0.9915
4,4'-BPF	Lufa c	17.9465	0.9016	0.9903	25.921	0.9744
BPA	Lufa d	27.0216	0.8999	0.9979	39.976	0.9800
BPS	Lufa d	10.8536	1.0136	0.9935	10.346	0.9881
2,2'-BPF	Lufa d	39.1291	1.1888	0.9976	19.538	0.9762
2,4'-BPF	Lufa d	46.6997	1.1904	0.9960	22.911	0.9786
4,4'-BPF	Lufa d	22.9542	1.1254	0.9993	14.647	0.9905

Table 5. Summary of model parameters of column experiments

No.	Soil	Organic carbon in % C	Contaminant	k_1(min^{-1})	k_{1d}(min^{-1})	k_2(min^{-1})	R^2
1	Lufa a	1.23	BPA	0.146	0.152	0.0310	0.98
2	Lufa a	1.23	BPS	0.326	0.332	0.0060	0.99
3	Lufa a	1.23	2,2'-BPF	0.253	0.321	0.0110	0.97
4	Lufa a	1.23	2,4'-BPF	0.212	0.221	0.0100	0.98
5	Lufa a	1.23	4,4'-BPF	0.558	0.654	0.0120	0.99
6	Lufa b	2.94	BPA	0.162	0.186	0.0670	0.99
7	Lufa b	2.94	BPS	0.488	0.502	0.0130	0.96
8	Lufa b	2.94	2,2'-BPF	0.123	0.135	0.0100	0.98
9	Lufa b	2.94	2,4'-BPF	0.320	0.453	0.0110	0.98
10	Lufa b	2.94	4,4'-BPF	0.201	0.213	0.0330	0.95
11	Lufa c	1.84	BPA	0.086	0.079	0.0440	0.99
12	Lufa c	1.84	BPS	0.563	0.662	0.0030	0.99
13	Lufa c	1.84	2,2'-BPF	0.103	0.181	0.0029	0.99
14	Lufa c	1.84	2,4'-BPF	0.580	0.652	0.0122	0.98
15	Lufa c	1.84	4,4'-BPF	0.462	0.533	0.0127	0.98
16	Lufa d	1.01	BPA	0.623	0.778	0.0300	0.98
17	Lufa d	1.01	BPS	0.801	0.963	0.0001	0.99
18	Lufa d	1.01	2,2'-BPF	0.832	0.866	0.0001	0.99
19	Lufa d	1.01	2,4'-BPF	0.632	0.733	0.0030	0.99
20	Lufa d	1.01	4,4'-BPF	0.825	0.901	0.0038	0.99

To further understand the transport of bisphenol analogs, batch adsorption experiments were performed under the same solution conditions as the column experiments. The results of batch adsorption experiment well explain the above column experiment phenomenon. The adsorption affinity of five bisphenol analogs with four typical soils was quantified by batch isothermal experiments. Through the adsorption model fitting, the Freundlich model fitting correlation coefficient $R^2 > 0.98$. It can better fit the adsorption isotherm data of bisphenol analogs on Lufa soils. The Freundlich model fitted well, indicating that bisphenol analogs formed multi-layer adsorption on the surface of Lufa soil. The fitting results of Freundlich model are shown in Tables 4 and 5. In the fitting results of Freundlich model, the value of 1/n between 0.75 and 1.25 indicates that the linear distribution plays a leading role, and the main medium of linear distribution is soil organic carbon. This result is consistent with the results of column experiments in which soil SOM content affects the transport of BPA and its substitutes. The K_f value reflects the adsorption affinity of bisphenol analogs to Lufa soil.

However, four different soils are used in batch adsorption experiments, different soils usually have different n values, and K_f can't be compared in different soils. Therefore, we use the approximate linear distribution coefficient (K_d) of the adsorption isotherm to compare the adsorption affinity of bisphenol analogs to soil. In four Lufa soils, for BPA, BPS and 4,4'-BPF, the K_d values in the four soils follow the following order: BPA > 4,4'-BPF > BPS. This suggests that the difference in sorption affinity of BPA, BPS and 4,4'-BPF in the four Lufa soils is the main factor contributing to the difference in their migration capacity.

Log K_{OW} is considered as a parameter to assess the hydrophobicity of organic chemicals (Christensen et al., 2022). The Log K_{OW} values of the three bisphenol analogs were ranked in the order of BPA > BPS > 4,4'-BPF. This was positively correlated with the K_d values of the three bisphenol analogs. The importance of hydrophobic interactions for the adsorption of bisphenol analogs has been reported in other studies (Choi &Lee, 2017; Guo et al., 2022; Wu et al., 2019). For example, (Wu et al., 2019) the adsorption of BPA, BPS, BPB and BPAF on PVC was found to have a good linear correlation with their hydrophobicity. In addition to hydrophobic interactions as described above, electrostatic and hydrogen bonding forces also affect the adsorption of bisphenol analogs by soil. In the experimental conditions of both column and batch adsorption experiments, the pH was close to 7, both below the pK_a of the bisphenol analogs, so they were in a non-separated state. Previous studies have demonstrated that the sorption of bisphenol analogs by soil does not change with pH when pH < pK_a (Choi &Lee., 2017). Therefore, electrostatic effects did not affect the adsorption of BPA and its substitutes in soil at the pH (6.7-7) of the batch adsorption experiment. In addition, hydrogen bonding forces are also a key reason for adsorption. BPA and its substitutes both contain hydroxyl, which readily forms hydrogen bonds with SOM in soil. It has been shown that BPS is more readily adsorbed on montmorillonite by Ca bridging and hydrogen bonding than BPA (Guo et al., 2022). However, among the four soils, for BPA and its isomers in soil adsorption affinity followed that BPA > BPA substitutes. Therefore, we judged that hydrophobic differences might be the main factor causing the difference in transport capacity of bisphenol analogs in the four standard soils.

3.2 Transport of 2,2'-BPF, 2,4'-BPF and 4,4'-BPF in saturated Lufa soils

Although 4,4'-BPF, 2,2'-BPF and 2,4'-BPF were almost all transported out of the soil column in soils Lufa a, Lufa c and Lufa d. Interestingly, they showed significant mobility differences in Lufa b soils with higher SOM content, where 2,2'-BPF and 2,4'-BPF were more mobility than 4,4'-BPF. The reason for this phenomenon may be that the three isomers of BPF -OH bonds are easily broken in the order of 4,4'-BPF < 2,2'-BPF < 2,4'-BPF. The broken hydroxyl will form hydrogen bonds with SOM in the soil, and more SOM in the soil will provide more adsorption sites for 4,4'-BPF and its two isomers (Wang et al., 2016). Therefore, the difference in mobility of 4,4'-BPF, 2,2'-BPF and 2,4'-BPF will probably become more significant as the SOM content in the soil increases. The results of the two-site kinetic retention model fitting showed that the values of the reversible adsorption site adsorption coefficient k_1 for 4,4'-BPF and its two isomers were greater than the irreversible adsorption site adsorption coefficient k_2. This indicates that more reversible adsorption of 4,4'-BPF and its two isomers occurred during the migration on the four Lufa soils due to their higher mobility and weaker affinity for soil adsorption.

The results of the batch adsorption experiments do not seem to explain well the results of the column experiments. From the adsorption isotherms (Fig. 4) and the adsorption data (Table 4), it can be seen that 2,4’-BPF and 2,2’-BPF have greater adsorption affinity for soil Lufa a, Lufa c and Lufa d than 4,4’-BPF. This clearly differs from the results of the column experiments, i.e., the differences in transport of the three BPF isomers in soil Lufa a, Lufa c and Lufa d were not significant. [(Pan et al., 2016)] showed that the hydrogen bond strength between the three isomers of BPF and 1-octyl-3-methylimidazole tetrafluoroborate ([Omim][BF4]) follows the following order: 2,2’-BPF > 2,4’-BPF > 4,4’-BPF. The SOM content of all three Lufa soils was low and the batch sorption experimental procedure lasted longer compared to the column experimental procedure, thus allowing the three isomers of BPF to more fully engage with these small amounts of SOM. Due to the ability of 2,2’-BPF and 2,4’-BPF to form stronger hydrogen bonds with SOM and bind more tightly to the soil, they exhibit a stronger affinity for soil adsorption than 4,4’-BPF. Based on the results of bulk sorption experiments and column experiments, it was shown that the three isomers of BPF differ in their sorption affinity for soil, and this difference is likely to become more significant as the SOM content increases. This may lead to their different fates and transport in the soil.

4. Conclusion

The results of batch adsorption experiments, column migration experiments and mathematical model fitting deepen the understanding of the retention and migration mechanisms of bisphenol analogs in natural soils. The use of ^{14}C-labeled bisphenol analogs and standard soils for column experiments and batch adsorption experiments can effectively improve the accuracy and reproducibility of experimental data. All experimental and fitting data showed that the transport capacity of BPA was higher than that of its substitutes BPS and 4,4’-BPF in the four different standard soils. In addition, the mobility of the three isomers of BPF may exhibit differences with increasing soil SOM content. The two-site kinetic model described well the experimental BTC of the bisphenol analogs, indicating that the model can be used to predict and detect the fate and migration of bisphenol analogs in soil. The experimental results suggest that the bisphenol alternatives BPS and 4,4’-BPF will pose a greater threat to groundwater. The higher mobility of 2,2'-BPF and 2,4'-BPF may pose a greater threat to groundwater than 4,4'-BPF, so the by-products 2,2'-BPF and 2,4'-BPF produced during BPF production should not be ignored. This study provides important insights into the fate and transport of bisphenol analogs in natural soils and helps to assess their potential environmental risks.

Declarations

Author’contribution

Shaoxin Zi: Investigation, Data curation, Writing—as-received draft. **Jiale Xu:** Investigation, Data curation. **Yingxin Zhang:** Investigation, Data curation. **Di Wu:** Writing—review & editing. **Jin Liu:** Conceptualization, Resources, Supervision, Writing—review & editing.

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Data availability

The datasets used during this study are available from the authors on reasonable request.

Compliance with ethical standards

Ethical approval Not applicable

Consent to participate Not applicable

Consent to Publish Not applicable

Competing interests The authors declare that they have no competing interests.

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Figures

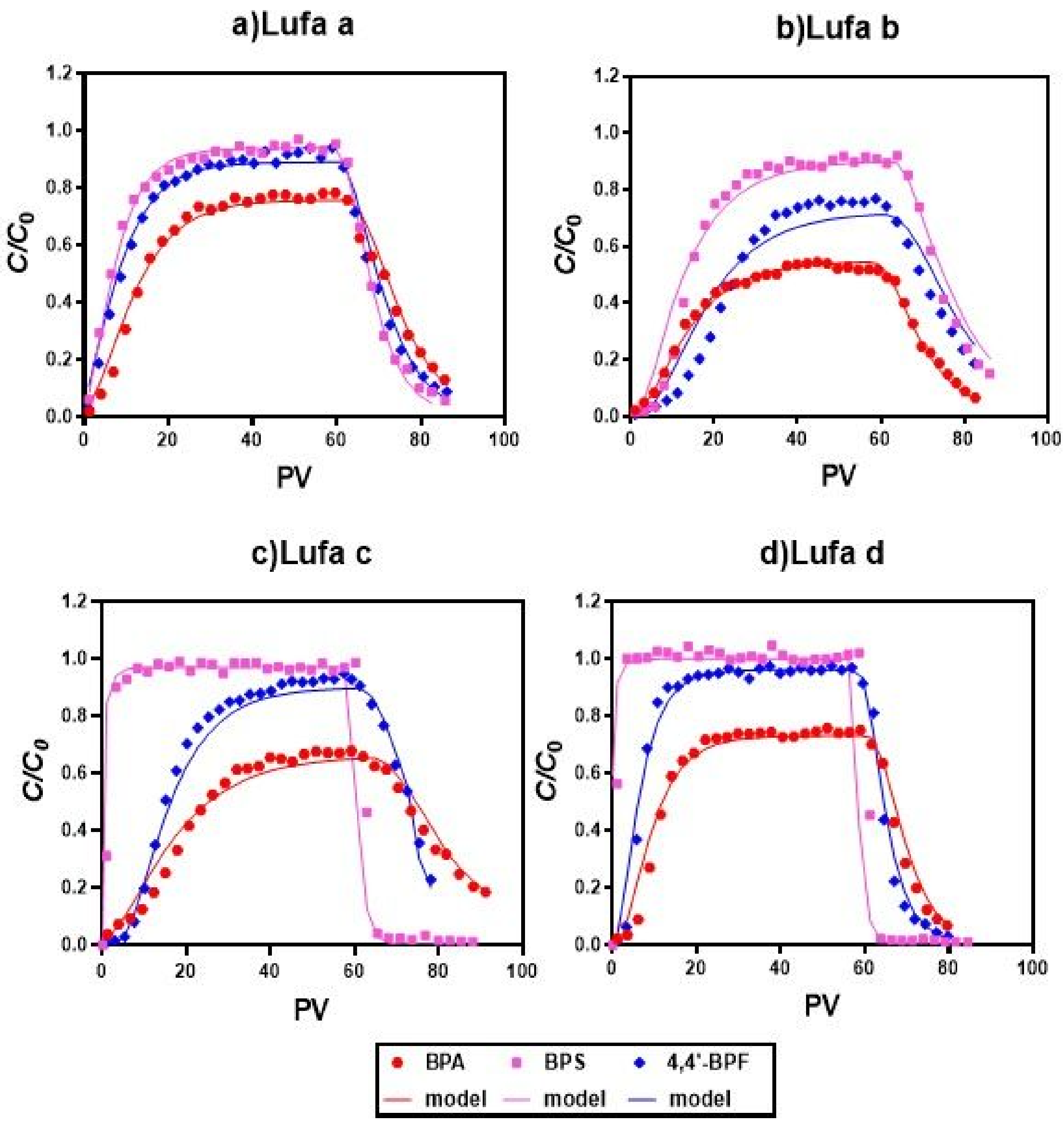

Figure 1

BTCs of BPA, BPS and 4,4’-BPF in saturated Lufa soils. Symbols are experimental data, and the lines are the fitted breakthrough curves using HYDRUS-1D.

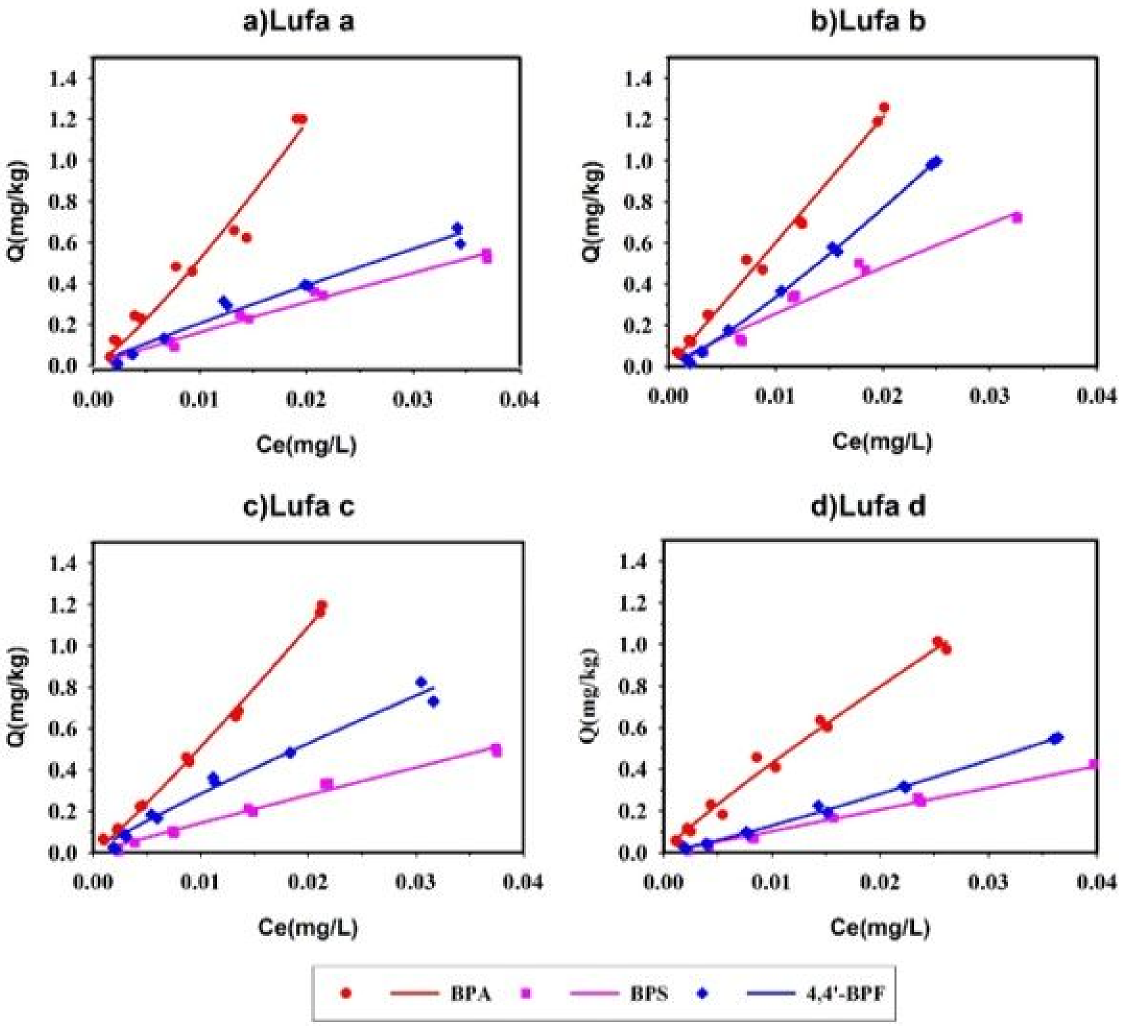

Figure 2

Adsorption isotherms of BPA, BPS and 4,4’-BPF to Lufa soils.

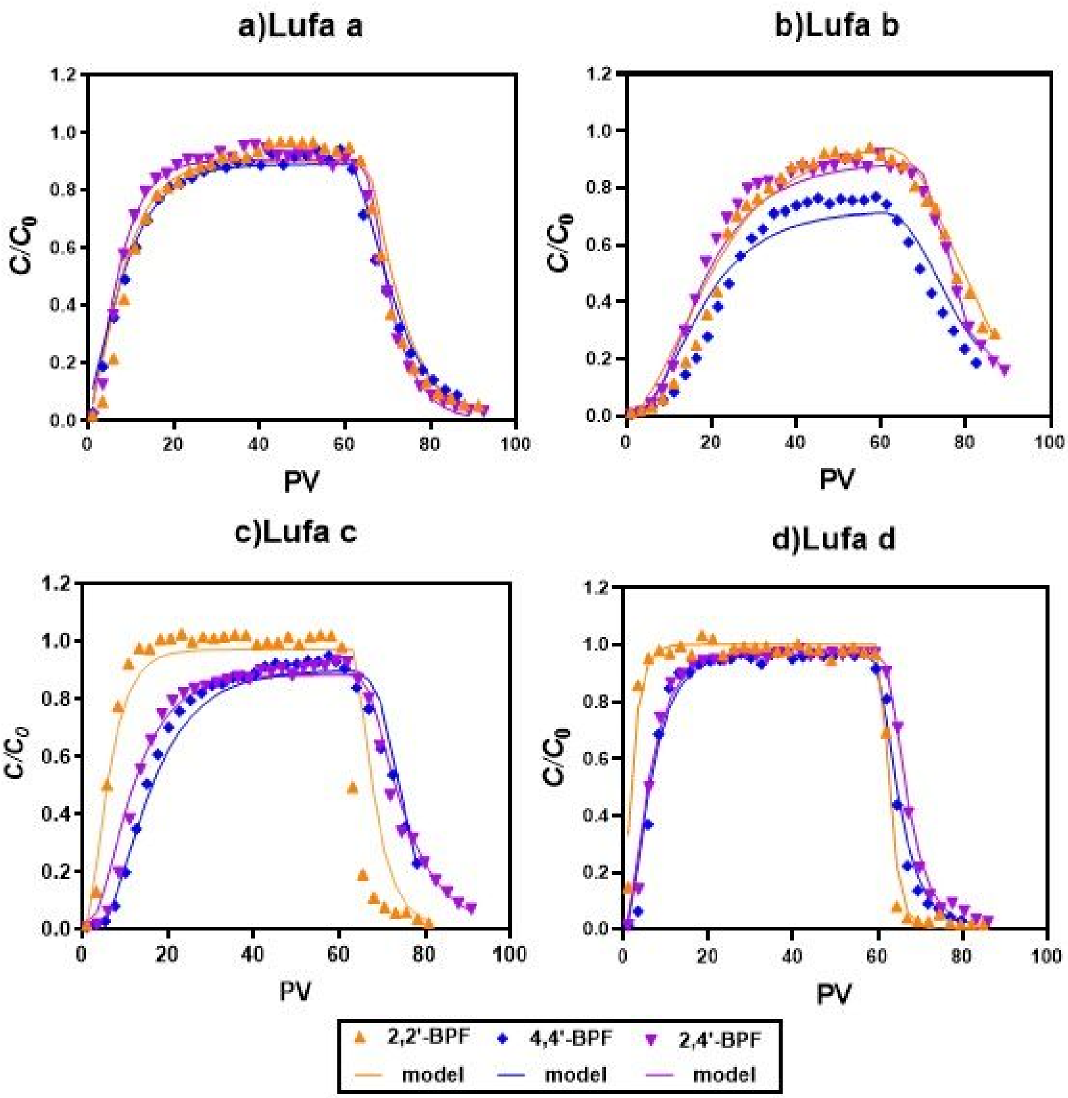

Figure 3

BTCs of 2,2’-BPF, 2,4’-BPF and 4,4’-BPF in saturated Lufa soils. Symbols are experimental data, and the lines are the fitted breakthrough curves using HYDRUS-1D.

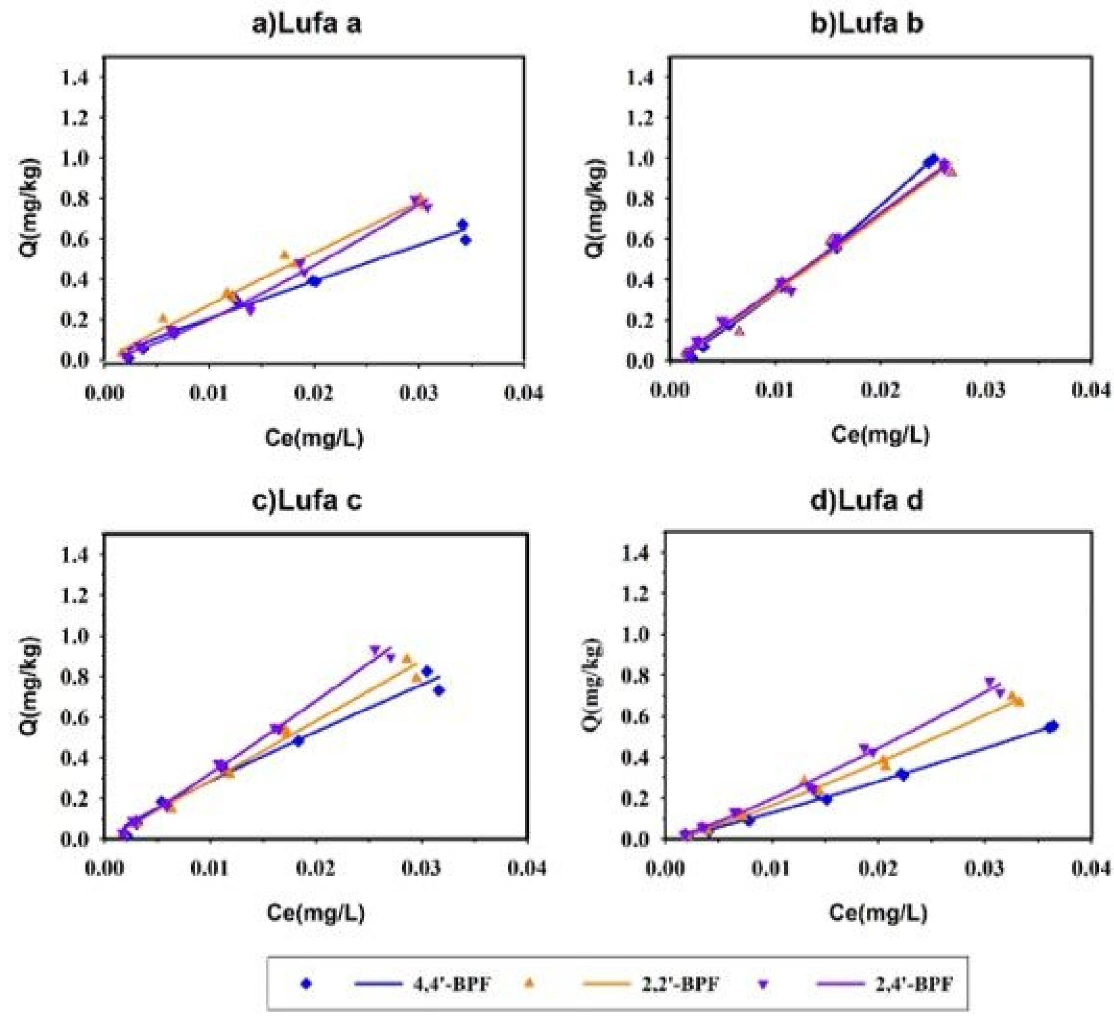

Figure 4

Adsorption isotherms of 2,2'-BPF, 2,4'-BPF and 4,4'-BPF to Lufa soils.